Quantum Mechanics as Coupled-Mode Theory: A Wave-Based Perspective on States, Operators, and Measurement

Warren Jackson

January 2026

Abstract

The formal structure of quantum mechanics includes unitary time evolution of a complex wavefunction governed by the Schrödinger equation with an explicit i and a separate, non-unitary postulate describing measurement and wavefunction collapse. This division obscures the physical origin of measurement and introduces apparent tension with relativistic causality, since collapse is often taken to occur instantaneously.

In this paper we show that much of the mathematical and conceptual structure of non-relativistic quantum mechanics—particularly measurement and collapse—can be understood within the framework of classical coupled-mode theory (CMT). CMT describes the interaction of discrete wave modes through overlap integrals and leads to evolution equations that are mathematically identical to the Schrödinger equation. Within this framework, wavefunctions correspond to mode-amplitude vectors, operators to physically realizable couplers, and inner products to spatial overlap integrals.

Measurement is modeled as unitary coupling between a system and auxiliary modes over a finite interaction length. Weak and strong measurements correspond to different coupling regimes, while strong measurements emerge as boundary-condition limits. Following the two-state formalism of Aharonov, the waveguide solution must statisfy a two-point boundary conditions. Measurement outcomes correspond to eigenstates because only self-consistent modes survive the interaction constraints. Apparent wavefunction collapse arises when auxiliary degrees of freedom—particularly their relative phases—are ignored, leading to effective mode selection rather than a breakdown of unitary evolution. Born probabilities emerge naturally as intensity partitioning among coupled modes.

Noncommutativity and canonical commutation relations are shown to originate from geometric phase accumulation associated with closed loops in phase space, while quantization follows from global phase self-consistency. No modification of quantum mechanics or new empirical predictions are proposed. Instead, the results provide a unified wave-based picture that embeds measurement within the same dynamical framework as ordinary evolution and removes the need for instantaneous collapse at the level of the physical description.

1. Introduction

Quantum mechanics is built on a remarkably compact and successful mathematical formalism. The Schrodinger equation provides a linear, unitary description of time evolution, and physical observables are represented by Hermitian operators acting on complex-valued wavefunctions. Within this framework, interference, superposition, and eigenvalue structure arise naturally.

Measurement, however, is treated differently. In the standard formulation, the unitary evolution governed by the Schrodinger equation is supplemented by a separate, non-unitary rule—often described as wavefunction collapse—according to which a measurement produces a definite outcome corresponding to an eigenstate of the measured operator. This rule is not derived from the dynamics but introduced as an additional postulate.

This separation raises both conceptual and physical difficulties. Collapse is not described as a dynamical interaction, yet it is triggered by physical measurement devices. Moreover, collapse is typically taken to occur instantaneously, a notion that sits uneasily with special relativity, where simultaneity is frame-dependent and no preferred time slicing exists. As a result, the standard measurement postulate appears incompatible with relativistic locality unless supplemented by additional interpretive structure.

At the same time, many of the mathematical features of quantum mechanics—complex amplitudes, explicit i, linear evolution, conserved norm, eigenvalue problems, and overlap-based transition amplitudes—are shared by classical wave systems. In particular, **classical coupled-mode theory (CMT)** provides a general description of how discrete wave modes interact through spatial overlap, leading to equations that are mathematically identical to the Schrodinger equation. This raises a natural question: *can measurement be understood as an ordinary wave interaction governed by the same principles as unitary evolution, rather than as an exceptional process?*

In this paper we show that much of the structure surrounding quantum measurement and collapse can be understood within the framework of coupled-mode theory. Measurement is modeled as unitary coupling between a system and auxiliary modes over a finite interaction length. Weak and strong measurements correspond to different coupling regimes, and strong measurements emerge as boundary-condition limits. Following the two-state representation of quantum mechanics by Aharonov, the waveguide functions must satisfy both an initial and final boundary conditions. This causes energy to become quantized. Measurement outcomes correspond to eigenstates because only self-consistent modes (eigenmodes) survive the interaction constraints. Apparent collapse arises when auxiliary degrees of freedom are not tracked, rather than from a breakdown of unitary dynamics.

Our aim is not to modify quantum mechanics or introduce new empirical predictions, but to provide a unified physical picture that embeds measurement within the same wave-based framework that governs ordinary evolution. This perspective clarifies which aspects of quantum measurement reflect general properties of interacting waves and which aspects are genuinely quantum.

2. Coupled-Mode Theory for Classical Waves

2.1 Modal expansion and complex amplitudes

Coupled-mode theory (CMT) provides a general framework for describing the interaction of waves in systems that support a discrete set of modes. It is widely used in optics, microwave engineering, acoustics, and mechanical resonators, and relies only on linear wave physics. In this section we review the essential elements of CMT needed for the subsequent comparison with quantum mechanics.

Consider a linear wave system whose fields can be expanded in a set of orthonormal transverse modes $\{\phi_n(\mathbf{r}_\perp)\}$. The total field may be written as

$$E(\mathbf{r}_\perp, z) = \sum_n a_n(z)\, \phi_n(\mathbf{r}_\perp),$$

where z is a distinguished evolution coordinate. In optical waveguides, z represents physical distance, while in resonators or lumped systems, it may represent time. The coefficients $a_n(z)$ are complex-valued modal amplitudes that encode both magnitude and phase.

The use of complex amplitudes is a standard feature of wave physics and reflects the oscillatory nature of solutions to linear wave equations. No probabilistic or quantum interpretation is implied at this stage.

These modes propagate along the waveguide with various frequencies and propagation velocities.

2.2 Inner product and mode normalization

The transverse modes solutions for waveguides are orthonormal under an inner product defined by an integral over the transverse coordinates,

$$\langle \phi_m \mid \phi_n \rangle = \int d\mathbf{r}_\perp\, \phi_m^*(\mathbf{r}_\perp)\, \phi_n(\mathbf{r}_\perp) = \delta_{mn}.$$

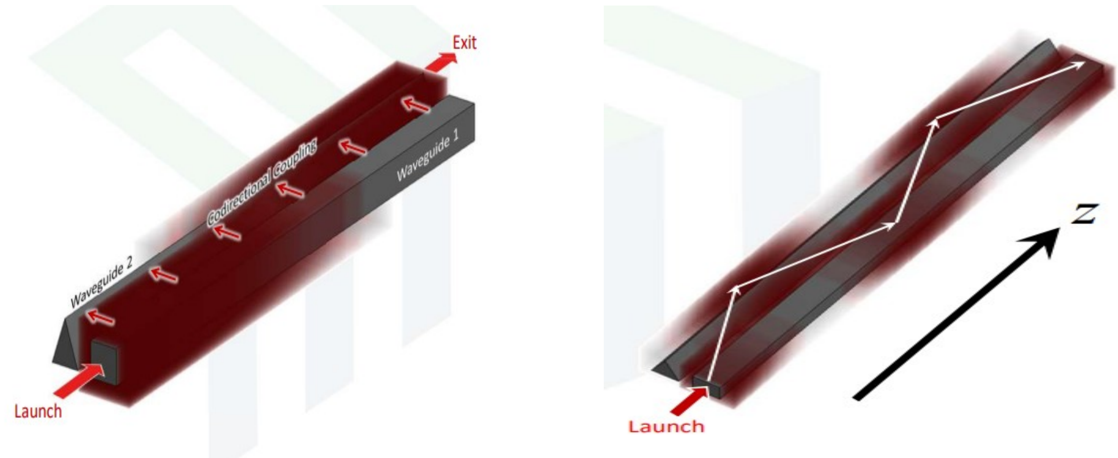
This inner product has a clear physical meaning: it measures the spatial overlap between modes. With this normalization, the total wave intensity or power carried by the field is proportional to

$$P = \sum_n |a_n(z)|^2.$$

In a closed, lossless system this quantity is conserved during evolution. This normalization by P even if P is small will later map directly onto probability conservation in quantum mechanics.

2.3 Mode coupling from physical interactions

When the system is perturbed—by interaction with another nearby waveguide, refractive-index variation, boundary deformation, or external modulation —the modes are no longer independent. Energy can be transferred coherently between them.



(a) (b)

Figure 1: (a) Two coupled waveguides coupled by effervescent field overlaps (b) Energy oscillates between the two modes for long interaction lengths

To first order, the evolution of the modal amplitudes is governed by

$$i\,\frac{d}{dz}a_m(z) = \sum_n H_{mn}\,a_n(z),$$

where the coupling matrix elements are given by overlap integrals

$$H_{mn} = \int d\mathbf{r}_\perp\,\phi_m^*(\mathbf{r}_\perp)\,\hat{H}(\mathbf{r}_\perp)\,\phi_n(\mathbf{r}_\perp).$$

Here $\hat{H}$ represents the physical interaction responsible for coupling between modes. The equation says that the rate at which one mode transfers amplitude to another is related to the local spatial overlap weighted by the interaction strength, a fact that will later explain the structure of quantum operators and inner products.

2.4 Hermiticity and conservation laws

For a reciprocal, lossless system, the coupling matrix H is Hermitian,

$$H_{mn} = H_{nm}^*.$$

Hermiticity guarantees conservation of total power,

$$\frac{d}{dz}\sum_n |a_n(z)|^2 = 0.$$

Thus, unitary evolution is not imposed by assumption but follows directly from energy conservation in classical wave systems. This point will later be central to understanding why quantum time evolution is unitary.

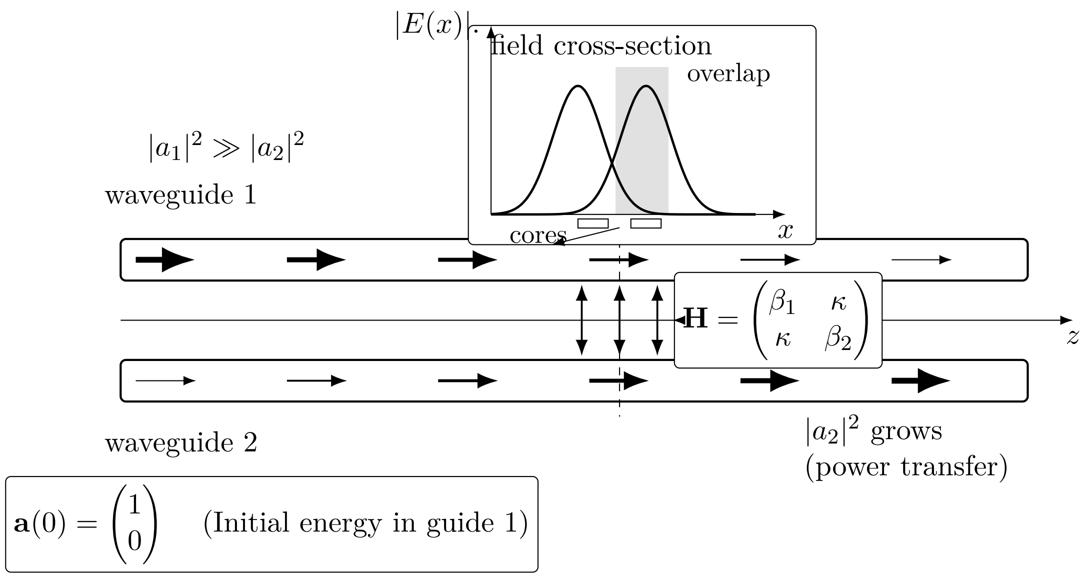


Figure 2: Two evanescently coupled waveguides. Power initially in waveguide 1 transfers to waveguide 2 via modal overlap. The local coupling is captured by the 2×2 coupled-mode matrix $\mathbf{H}$.

2.5 Normal modes and eigenvalue structure

The eigenvectors of the coupling matrix H correspond to the normal modes of the coupled system. If the system is prepared in one of these eigenmodes, it propagates without changing shape, accumulating only a phase. Superpositions of eigenmodes generally evolve nontrivially.

This eigenvalue structure is a generic feature of interacting wave systems and does not depend on quantum mechanics. As we shall see, the same structure underlies energy eigenstates, measurement eigenstates, and operator spectra in quantum theory.

2.6 Scope and relevance for quantum mechanics

The coupled-mode equation derived here has the same mathematical form as the Schrodinger equation. The appearance of complex amplitudes, linear evolution, Hermitian generators, inner products defined by spatial integrals, and conserved norm all arise from classical wave physics alone.

In the sections that follow, this correspondence will be used to interpret quantum wavefunctions, operators, measurement, and commutation relations in concrete physical terms. The purpose is not to reduce quantum mechanics to classical mechanics, but to show that much of its formal structure reflects general properties of waves interacting with waves.

3. The Schrodinger Equation as a Coupled-Mode Equation

One of the most striking features of quantum mechanics is the specific form of the Schrodinger equation: a linear, first order differential equation in time with complex-valued amplitudes, a Hermitian generator and an explicit presence of i which converts a first order rate equation into a wave equation. In standard presentations, this structure is introduced axiomatically, often without physical motivation. In this section, we show that this form arises naturally—and essentially uniquely—from classical coupled-mode theory (CMT), including the origin of the Hamiltonian matrix elements and the quantum-mechanical inner product.

3.1 Identification with Schrodinger time evolution

The formal correspondence with quantum mechanics becomes immediate upon identifying the propagation coordinate z with time t, and the modal amplitude vector $\mathbf{a}$ with a quantum state vector $|\psi\rangle$. The coupled-mode equation then becomes

$$i\,\frac{d}{dt}|\psi(t)\rangle = H\,|\psi(t)\rangle,$$

which is precisely the Schrodinger equation (in units where $\hbar = 1$, or with $\hbar$ absorbed into the definition of H).

In this identification, the Hamiltonian matrix elements

$$\langle \phi_m | H | \phi_n \rangle$$

are not abstract postulates, but explicit overlap integrals over the degrees of freedom transverse to the evolution coordinate. The Schrodinger equation thus appears to be a coupled-mode equation describing the coherent interaction of waves. The Hamiltonian shows how the amplitude from one wavefunction is transferred to other modes analogous to interacting waveguides exchanging excitation. In both cases„ the interaction is proportional to the transverse spatial overlap weighted by the interaction. The complex wave function is essential to represent the wave amplitude. The unique first order time derivative in both equations arise because they represent first order rate equations of the form

$$\frac{d}{dt}N = \mathbf{R}N$$

for waves rather than for scalars.

3.2 Origin of the quantum inner product

This correspondence provides a direct physical interpretation of the quantum-mechanical inner product. In CMT, the natural inner product between two modes ϕ and ψ is defined as

$$\langle \phi | \psi \rangle = \int d\mathbf{r}_\perp\, \phi^*(\mathbf{r}_\perp)\, \psi(\mathbf{r}_\perp),$$

an overlap integral over the transverse spatial dimensions.

The same structure appears in quantum mechanics, where the inner product between wavefunctions is defined as an integral over 3-d space. In the present view, this is not a mysterious probabilistic construction but a direct inheritance from classical wave theory: inner products arise because coupling strengths depend on local spatial overlap.

Thus, expressions of the form

$$\langle\phi|A|\psi\rangle$$

are naturally interpreted as overlap integrals of one wave with another, weighted by an interaction operator A. Quantum expectation values and transition amplitudes are therefore understood as physically meaningful measures of how strongly one wave configuration couples to another. In QM, it is usual to mysteriously 'promote' classical quantities such as $\boldsymbol{E}$, $\boldsymbol{x}$, and $\boldsymbol{p}$ to operators. This process is understood to represent these quantities as couplings between a waveguide representing the wave function $|\psi\rangle$ before the application of the operator (Waveguide 1 in Figure 2) and Waveguide 2 as the result $H|\psi\rangle$ of the application of operator $\boldsymbol{H}$.

3.3 Why the equation must be first order and complex

Within the CMT framework, the structure of the Schrodinger equation becomes clear. The Schrodinger Eq. represents a coupling between various eigenmodes of the waveguide. The rate at which energy, analogous to probability, transitions between modes is proportional to the overlap and the coupling strength. The first-order evolution equation propagates phase information directly and preserves the norm of the amplitude vector. A second-order equation by contrast, would require additional initial conditions and would not naturally conserve a simple quadratic norm.

The explicit appearance of the imaginary unit i is likewise unavoidable. In coupled-mode theory, coupling between modes produces phase rotation rather than exponential growth or decay. The factor of i encodes oscillatory exchange of energy between modes and ensures conservation of energy. The same logic applies in quantum mechanics: without i, time evolution would not be unitary.

3.4 Hermiticity and norm conservation

Hermiticity of the coupling matrix H follows directly from the symmetry of the overlap integrals and guarantees conservation of total wave intensity,

$$\frac{d}{dz}\left(\mathbf{a}^\dagger\mathbf{a}\right) = 0.$$

Under the Schrodinger correspondence, this becomes conservation of total probability,

$$\frac{d}{dt}\langle\psi|\psi\rangle = 0.$$

From this perspective, probability conservation is simply the statement that energy or amplitude is neither created nor destroyed in a closed, lossless wave system.

3.5 Physical meaning of the Hamiltonian

The Hamiltonian operator thus acquires a clear physical interpretation. Its diagonal elements represent frequency of the mode along the waveguide i.e. the phase accumulation of individual modes, while its off-diagonal elements represent coherent coupling between modes through spatial overlap. Energy eigenstates correspond to normal modes of the coupled system and energy eigenvalues correspond to their associated propagation constants.

This interpretation removes much of the abstraction traditionally associated with the Hamiltonian. It is not a uniquely quantum object, but a compact representation of how waves interact with waves.

3.6 Summary

The Schrodinger equation emerges naturally from classical coupled-mode theory once wave amplitudes are treated as complex quantities and norm conservation is imposed. Its linear structure, first-order time dependence, Hermitian generator, and inner-product formalism all follow directly from the physics of interacting wave modes. This correspondence provides a concrete physical foundation for the quantum formalism and prepares the way for similarly transparent interpretations of operators and measurement.

4. Wavefunctions, Operators, and Tensor Products as Physical Objects

The formalism of quantum mechanics is often introduced in a highly abstract way: wavefunctions are elements of a Hilbert space, operators act on these elements, and composite systems are described by tensor products. While mathematically precise, this presentation can obscure the physical meaning of these objects. In this section, we reinterpret wavefunctions, operators, and tensor products in the concrete language of coupled-mode theory (CMT), where each acquires a direct physical realization.

4.1 The wavefunction as a mode-amplitude vector

In coupled-mode theory, a wave propagating in a structured medium is expanded in a set of orthonormal transverse modes $\{\phi_n(\mathbf{r}_\perp)\}$. The state of the system is fully specified by the complex amplitudes $\{a_n\}$, which encode both the magnitude and phase of each mode. These amplitudes evolve according to the coupled-mode equation derived in Section 3.

This mode-amplitude vector is directly analogous to the quantum wavefunction $|\psi\rangle$. Rather than representing probability itself, the wavefunction represents the state of excitation of a set of modes. Probabilistic interpretation enters only when measurement is introduced, not at the level of wave propagation.

Thus, in the present picture, a quantum state is most naturally viewed as a classical wave state, expressed in an abstract mode basis rather than in physical space.

4.2 Operators as physical couplers

An operator A acting on a quantum state $|\psi\rangle$ produces a new state $A|\psi\rangle$. In the coupled-mode picture, this operation has a simple interpretation: it corresponds to coupling the system waveguide to an auxiliary waveguide whose modal amplitudes represent the result of the operation.

More explicitly, if $|\psi\rangle$ has components ψ_n in a chosen basis, then

$$(A|\psi\rangle)_m = \sum_n A_{mn}\,\psi_n,$$

where the matrix elements A_{mn} are overlap integrals of the form discussed in Section 3. Physically, A specifies how the amplitude is redistributed among modes by an interaction.

From this perspective:

- diagonal operators correspond to phase shifts of individual modes along the waveguide;
- off-diagonal operators correspond to coherent coupling between modes;
- Hermitian operators correspond to lossless, reversible couplings.

This interpretation removes much of the mystery surrounding quantum operators: they are not abstract instructions, but compact representations of physically realizable coupler networks.

4.3 Position and momentum as special couplers

Within this framework, the familiar position and momentum operators acquire particularly clear meanings. The position operator acts as a phase-gradient coupler, imposing a spatially dependent phase across the waveguide. The momentum operator acts as a mode-mixing coupler, redistributing amplitude among neighboring spatial modes.

These two operators generate translations in complementary representations and, as discussed later, their noncommutativity reflects the fact that phase gradients and spatial redistribution do not commute. Importantly, both operators correspond to physically realizable wave interactions and do not require uniquely quantum assumptions.

4.4 The tensor product as a coupled-mode construction

The tensor product is often introduced as a purely mathematical operation used to describe composite systems. In the coupled-mode picture, however, its physical meaning is straightforward.

Consider two independent waveguides, each supporting its own set of modes. The combined system supports modes that are products of the individual modes. If the first system has modes $\{\phi_i\}$ and the second has modes $\{\chi_j\}$, then the combined system has modes $\{\phi_i\chi_j\}$. The amplitude associated with each product mode specifies how much excitation resides simultaneously in mode i of the first system and mode j of the second.

This is precisely the tensor product construction,

$$|\psi\rangle_{\text{total}} = |\psi\rangle_1 \otimes |\psi\rangle_2.$$

Thus, the tensor product space is simply the **mode space of the combined wave system**.

4.5 Operators on composite systems

Operators acting on composite systems also acquire a transparent meaning. An operator of the form

$$A \otimes I$$

acts only on the modes of the first waveguide, leaving the second unchanged. Similarly,

$$I \otimes B$$

acts only on the second waveguide. Interactions between the two systems correspond to operators with nontrivial matrix elements coupling product modes, and therefore to physical couplers linking the two waveguides.

Entanglement arises naturally when the coupling mixes product modes in such a way that the total wave cannot be written as a simple product of individual states. In this case, the excitation resides in correlated mode combinations across the two systems, exactly as in coupled waveguides whose normal modes are delocalized across both structures.

4.6 Expectation values and transition amplitudes

Within this picture, expressions such as

$$\langle A \rangle = \langle \psi | A | \psi \rangle$$

are interpreted as overlap integrals measuring how strongly a forward time wave couples to its backwards time conjugate under the action of A from the two state interpretation of Similarly, transition amplitudes the weak measurement

$$\langle A \rangle_w = \langle \phi | A | \psi \rangle / \langle \phi | \psi \rangle$$

measure the strength of coupling between different forward time modes and backward time modes normalized by the overlap. Thus, it is a measure of the coupling strength with the effect of overlap removed for a given cross section of the waveguide.

These quantities are not abstract probabilities but physically meaningful measures of interaction strength, inherited directly from classical wave theory.

4.7 Summary

Wavefunctions, operators, and tensor products acquire concrete physical interpretations when viewed through the lens of coupled-mode theory. A wavefunction is a vector of modal amplitudes, an operator is a structured coupler that redistributes those amplitudes, and a tensor product is the natural mode space of a composite wave system. This perspective demystifies the quantum formalism and prepares the ground for understanding measurement as a physical interaction rather than as an external postulate.

5. Measurement as Unitary Coupling and Effective Mode Selection

In standard formulations of quantum mechanics, measurement is introduced as a non-unitary projection that interrupts otherwise unitary evolution. In the coupled-mode framework developed here, measurement is instead modeled as unitary coupling between a system and an auxiliary waveguide . Apparent non-unitarity—and what is traditionally called wavefunction collapse—arises only when the auxiliary degrees of freedom are not tracked in detail.

5.1 Measurement as coupling to auxiliary modes

Consider a system waveguide carrying a state $|\psi\rangle$, coupled to a set of auxiliary waveguides representing a measurement apparatus. The interaction is characterized by a coupling strength κ and occurs over a finite interaction length L.

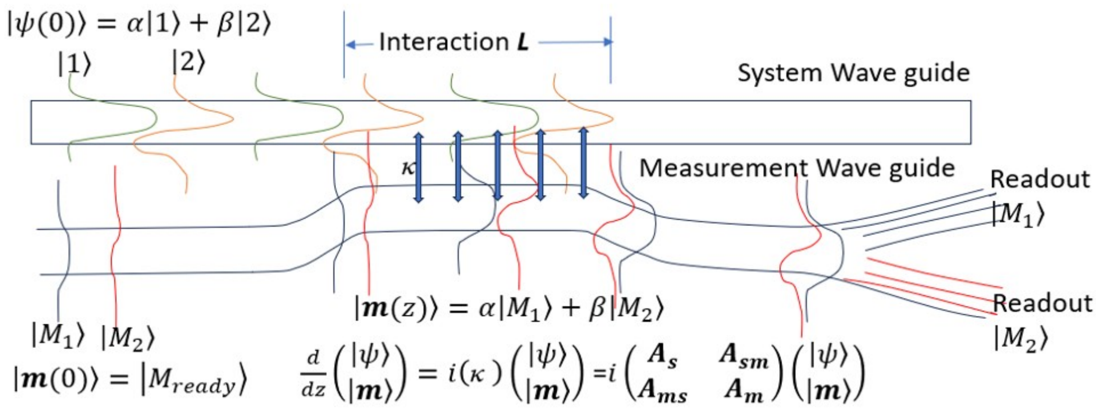


Figure 3: Measurement by coupling between the system waveguide and the measurement waveguide over an interaction length $\boldsymbol{L}$ and coupling κ

The combined system–auxiliary state $|\Psi\rangle$ evolves according to a unitary Schrodinger equation,

$$i\frac{d}{dt}|\Psi\rangle = (H_{\text{sys}} + H_{\text{aux}} + H_{\text{int}})\,|\Psi\rangle,$$

where H_{int} produces coherent transfer of amplitude between system modes and auxiliary modes. At no point is non-unitary dynamics required.

5.2 Correlation and phase entanglement

If the system is prepared in a superposition of eigenstates of the measured operator,

$$|\psi\rangle = \sum_n c_n|\phi_n\rangle,$$

then the interaction produces a correlated state of the form

$$|\Psi\rangle = \sum_n c_n|\phi_n\rangle \otimes |m_n\rangle,$$

where $|m_n\rangle$ denotes auxiliary modes associated with outcome n.

Crucially, each auxiliary mode carries its own phase evolution. The system and auxiliary degrees of freedom are therefore not only correlated in amplitude but also entangled in phase. This phase information is distributed across many auxiliary degrees of freedom.

5.3 Weak and strong measurement revisited

The character of the measurement interaction is controlled by the dimensionless parameter κL:

- **Weak measurement** ($\kappa L \ll 1$): Only a small fraction of the system amplitude couples into the auxiliary modes. The auxiliary phases remain only weakly correlated with the system, and interference between system components is largely preserved.

- **Strong measurement** ($\kappa L \gg 1$): The system becomes strongly correlated with auxiliary measurement modes over the interaction region. The system wavefunction is destroyed. Phase information associated with different system components is rapidly distributed into distinct measurement auxiliary channels each corresponding to a measurement value. The modes which propagate in the measurement waveguide must be eigenfunctions of the measurement waveguide. Hence, only eigenvalues of are possible measurements although at this point they could be linear combinations of the eigenvalues.

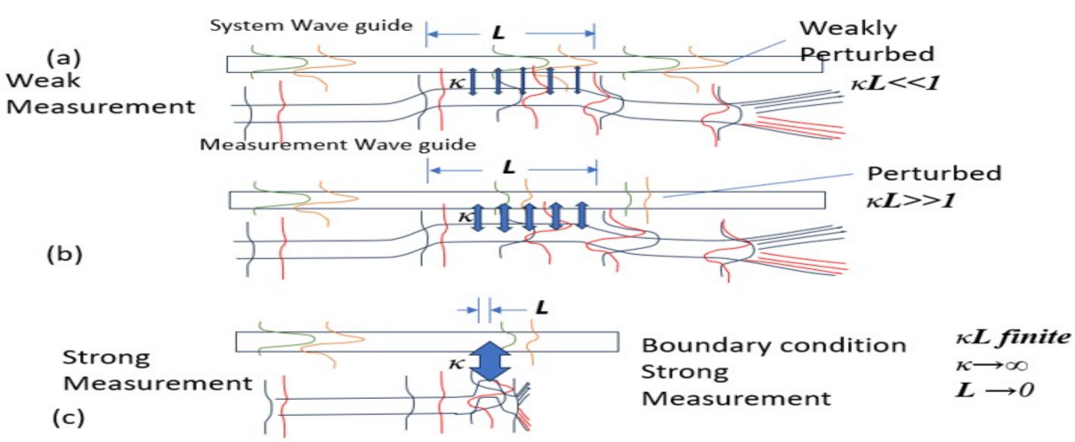


Figure 4: Weak and Strong measurement depend on the coupling strength κ and interaction length L. The ability to resolve the measurement also depends on L.

In the limit $L \to 0$, $\kappa \to \infty$ with κL finite, the interaction acts as a boundary condition imposed at a single propagation coordinate z for a waveguide or t in quantum mechanics. The strong measurement destroys the system mode and acts like a filter for the eigenfunctions representing the possible measurement eigenvalues .

By usual Fourier series properties, the uncertainty in the waveguide variable z, Δz and the uncertainty of the measurement amplitude has the relationship

$$(\Delta z)^2 (\Delta A)^2 > 2\pi,$$

Hence, the uncertainty relationship in QM between energy and time is explained by the interaction time.

5.4 Ignoring auxiliary phases and effective non-unitarity

In realistic measurement situations, the auxiliary degrees of freedom are not monitored in detail. In particular, their relative phases are not tracked. From the coupled-mode perspective, this corresponds to coarse-graining over many auxiliary modes with effectively uncontrollable or rapidly varying phases shown by transition from a few measurement waveguide modes to modes corresponding to different eigenmodes of the measurement waveguide.

When this phase information is ignored, interference terms between different components of the system wavefunction no longer contribute to observable quantities. Mathematically, this corresponds to tracing over auxiliary degrees of freedom; physically, it corresponds to losing access to phase coherence stored outside the system waveguide.

At this stage, the system no longer behaves as if it were in a coherent superposition. Instead, it appears to occupy a single mode drawn from a set of self-consistent outcomes.

5.5 Collapse as effective mode selection

What is traditionally called wavefunction collapse is therefore reinterpreted as effective mode selection:

- The full system–auxiliary evolution remains unitary.
- Multiple correlated system–auxiliary branches coexist at the level of the combined wave.
- When auxiliary phases are ignored, only one system mode remains dynamically self-consistent and observable.

Collapse is not a physical discontinuity in time, but a consequence of restricting attention to a subset of the total wave system. The apparent non-unitarity reflects a change in description, not a change in dynamics.

5.6 Probabilities and intensity partitioning

The coefficients $|c_n|^2$ determine how much wave intensity is transferred into each auxiliary channel. When auxiliary phases are ignored, these intensities behave exactly like classical probabilities.

Thus, the Born rule emerges as a statement about intensity partitioning among coupled waveguides, not as a fundamental probabilistic axiom. Probability reflects how wave energy is distributed when phase information is unavailable.

5.7 Summary

Measurement is a unitary coupling process that correlates system modes with auxiliary modes and distributes phase information across degrees of freedom that are typically not tracked. Measurements are identical to other quantum mechanics interactions. Apparent dissipation is leakage into other modes rather than some novel nonunitary interaction. Weak and strong measurements form a continuum and correspond to different coupling regimes, while strong, localized interactions act as boundary conditions. Apparent collapse arises when auxiliary phases are ignored, leading to effective mode selection rather than a breakdown of unitary evolution.

6. Conjugate Variables

If operators are couplings between modes, then what are conjugate variable operators in the CMT frame work. In particular consider X and P. We find that the conjugate variables are quadratures of a waveguide mode.

6.1 Conjugate variables as quadratures of a wave

The operators X and P play a special role because they form a quadrature pair. Just as the real and imaginary parts of a complex wave amplitude describe orthogonal components of a single oscillation, position and momentum represent complementary descriptions of the same underlying wave.

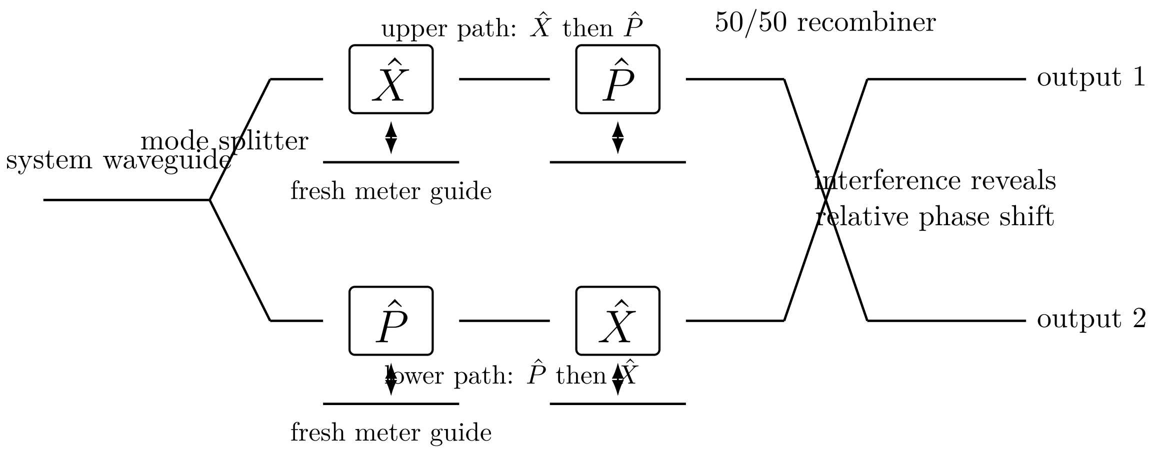


Figure 5: Interferometric demonstration of operator ordering using coupled-waveguide interactions. A system waveguide is split into two paths. In the upper arm the system interacts with a position-type coupling $\hat{X}$ followed by a momentum-type coupling $\hat{P}$, each using fresh ancillary guides. In the lower arm the order is reversed. The two paths are recombined in a 50/50 coupler, allowing interference to reveal any relative phase shift associated with the non-commuting ordering of $\hat{X}$ and $\hat{P}$ interactions.

In the coupled-mode framework:

- X acts as a phase-gradient operator, imposing a spatially dependent phase across the wavefunction;
- P acts as a mode-mixing operator, redistributing amplitude among neighboring spatial modes.

These two operations correspond to orthogonal quadratures of wave evolution. Fixing one necessarily delocalizes the other, reflecting a purely wave-based constraint rather than a uniquely quantum assumption.

6.2 Generators of translations as physical couplers

A second defining feature of X and P is that they generate translations:

- X generates translations in momentum,
- P generates translations in position.

In the coupled-mode picture, these translations are implemented by distinct physical couplers. A phase-gradient coupler and a mode-mixing coupler act in fundamentally different ways on the wave, even though both are linear and unitary. Each is physically realizable in classical wave systems, for example through phase masks, graded-index media, diffraction, or evanescent coupling.

Because these couplers represent different physical processes, their effects do not commute.

6.3 Noncommutativity as order-dependent coupling

Consider applying the two couplers in different orders. Applying a phase gradient and then redistributing amplitude does not, in general, produce the same result as redistributing amplitude first and then applying the phase gradient.

In coupled-mode theory, this order dependence has a simple interpretation: phase accumulation and amplitude redistribution are not interchangeable operations. When the two operations are applied sequentially and then reversed, the system returns to its original mode configuration but acquires an additional global phase.

This residual phase is the physical content of the commutator. Noncommutativity reflects not a failure of classical logic, but the fact that closed sequences of wave interactions do not return a wave to itself exactly, only up to a phase.

6.4 Commutators and phase-space loops

The phase acquired under a closed sequence of noncommuting couplers is a geometric phase. It depends only on the path taken in the space of transformations, not on the detailed timing of the operations.

For conjugate variables such as X and P, the relevant space is phase space. Alternating infinitesimal translations in position and momentum trace out a closed loop in phase space. The accumulated phase is proportional to the area enclosed by the loop.

The canonical commutation relation

$$[X, P] = i\hbar$$

may therefore be interpreted as a local statement: each infinitesimal phase-space loop contributes a fixed phase proportional to its area. In this sense, $\hbar$ acts as a conversion factor between phase-space area and phase.

6.5 Action quantization and global phase consistency

Requiring that wave evolution be globally self-consistent imposes an additional constraint. When a wave is transported around a closed loop in phase space, it must reproduce itself not only in amplitude but also in phase. This requirement restricts the allowed loops to those whose accumulated phase is an integer multiple of 2π.

As a result, the enclosed phase-space area—and hence the action—is quantized. This condition provides a natural explanation for discrete energy levels and the appearance of ladder-operator structure.

Creation and annihilation operators correspond to operations that increase or decrease the area of a self-consistent phase-space loop by one quantum. In this way, multiparticle states and excitation number emerge directly from phase-space geometry rather than from discretization imposed at the outset.

6.6 What is wave-geometric and what is quantum

The coupled-mode picture clarifies which aspects of noncommutativity are generic wave phenomena and which are genuinely quantum:

- noncommuting operations and geometric phase arise in classical wave systems;
- the restriction to globally phase-consistent loops is a quantum constraint.

Quantum mechanics does not introduce noncommutativity *ex nihilo*; it restricts which wave-geometric structures are physically admissible.

6.7 Summary

Conjugate variables arise as quadrature components of wave evolution and act as generators of distinct physical couplings. Their noncommutativity reflects geometric phase accumulation under closed phase-space loops. Canonical commutation relations encode the local curvature of phase space, while quantization emerges from the requirement of global phase self-consistency. In this way, commutators, uncertainty relations, and ladder-operator structure follow directly from wave dynamics rather than from uniquely quantum postulates.

7. Conclusion

We have shown that the core structures associated with quantum measurement—eigenstate outcomes, state reduction, and measurement-induced uncertainty—arise naturally when wave evolution is described using coupled-mode theory and subject to interaction and boundary constraints. The Schrodinger equation appears as the generic evolution equation for interacting wave modes with conserved norm, while operators correspond to physically realizable couplers whose matrix elements are determined by overlap integrals.

Within this framework, measurement is not a special, non-dynamical process. It is a unitary interaction between a system and auxiliary modes occurring over a finite interaction length. Weak and strong measurements differ only in coupling strength and duration, and strong measurements correspond to boundary-condition limits. Measurement outcomes are eigenstates because only those modes remain self-consistent under the imposed interaction constraints. What is traditionally described as wavefunction collapse is reinterpreted as effective mode selection when auxiliary degrees of freedom are ignored.

Noncommutativity and canonical commutation relations emerge from the geometry of wave coupling in phase space. Conjugate variables act as quadrature components of wave evolution, and their commutators encode geometric phase accumulation around closed phase-space loops. Quantization of action follows from the requirement of global phase self-consistency rather than from discretization imposed by postulate.

Taken together, these results suggest that many features often regarded as uniquely quantum reflect general properties of interacting waves, while the genuinely quantum aspect lies in the restriction to globally phase space self-consistent solutions. By embedding measurement within unitary wave dynamics, this approach avoids the need for instantaneous collapse and removes an apparent incompatibility with relativistic causality at the level of the dynamical description.

Broader implications of this perspective—particularly those related to time symmetry, retro-consistency, and quantum correlations—are beyond the scope of the present paper and will be addressed separately. Here, the focus has been on establishing a clear and physically grounded correspondence between coupled-mode theory and the formal structure of quantum mechanics.

References

[1] H. A. Haus, *Waves and Fields in Optoelectronics* (Prentice Hall, 1984).

[2] A. Yariv, "Coupled-mode theory for guided-wave optics," *IEEE J. Quantum Electron.* **QE-9**, 919–933 (1973).

[3] D. Marcuse, *Theory of Dielectric Optical Waveguides* (Academic Press, 1991).

[4] J. J. Sakurai and J. Napolitano, *Modern Quantum Mechanics*, 2nd ed. (Addison–Wesley, 2011).

[5] P. A. M. Dirac, *The Principles of Quantum Mechanics* (Oxford University Press, 1930).

[6] J. von Neumann, *Mathematical Foundations of Quantum Mechanics* (Princeton University Press, 1955).

[7] Y. Aharonov, D. Z. Albert, and L. Vaidman, "How the result of a measurement of a component of the spin of a spin-$\frac{1}{2}$ particle can turn out to be 100," *Phys. Rev. Lett.* **60**, 1351–1354 (1988).

[8] Y. Aharonov and L. Vaidman, "Properties of a quantum system during the time interval between two measurements," *Phys. Rev. A* **41**, 11–20 (1990).

[9] H. D. Zeh, "On the interpretation of measurement in quantum theory," *Found. Phys.* **1**, 69–76 (1970).

[10] W. H. Zurek, "Decoherence and the transition from quantum to classical," *Phys. Today* **44**(10), 36–44 (1991).

[11] M. Schlosshauer, *Decoherence and the Quantum-to-Classical Transition* (Springer, 2007).

[12] A. E. Siegman, *Lasers* (University Science Books, 1986).

[13] H. Haken, *Light: Laser Light Dynamics* (North-Holland, 1985).

[14] M. V. Berry, "Quantal phase factors accompanying adiabatic changes," *Proc. R. Soc. Lond. A* **392**, 45–57 (1984).

[15] A. Bohm, A. Mostafazadeh, H. Koizumi, Q. Niu, and J. Zwanziger, *The Geometric Phase in Quantum Systems* (Springer, 2003).

[16] R. G. Littlejohn, "The semiclassical evolution of wave packets," *Phys. Rep.* **138**, 193–291 (1986).

[17] L. D. Landau and E. M. Lifshitz, *Quantum Mechanics: Non-Relativistic Theory* (Pergamon, 1977).

[18] J. S. Bell, *Speakable and Unspeakable in Quantum Mechanics* (Cambridge University Press, 1987).